

Fixed-Spectrum Attention Routing for Length-Generalizing Retrieval

Anonymous submission

Abstract

Transformers can learn retrieval rules at short context lengths and still fail as competing tokens accumulate. Attention concentration is often part of the story, yet the same spectrum can place its large weights on useful evidence or irrelevant context. We therefore separate a spectrum’s selective capacity from its assignment and from the downstream utility of the values it moves. Spectrum-preserving swaps let us test that separation on controlled retrieval, pretrained models, and natural question answering. They show when a reassignment changes the prediction, and they also expose saturation and selector mismatch. The result is a causal assay for attention routing in length-generalizing retrieval.

1 Background

A transformer may learn a retrieval rule on short sequences and then lose it as distractors accumulate. We see this pattern in copying, key–value recall, and algorithmic tasks (Kazemnejad et al. 2023; Wang et al. 2024; Vasylenko et al. 2025). As contexts grow, attention often becomes more diffuse, entropy rises, and attention-output variance can collapse (Li, Boduljak, and Zhou 2025). Those trends describe scale and concentration, yet they do not tell us where the attention goes.

That gap is structural. Two attention rows can share their sorted weights, entropy, maximum, and every norm while placing their largest weights on different tokens. One can favor the evidence and the other a distractor. This leads to our first question: *At fixed concentration, does assigning the same attention spectrum to task-required evidence change the prediction?*

We answer with a fixed-spectrum intervention. At the answer query, we exchange the weight on a known source with a distractor weight and then run the unchanged model. Because the exchange preserves the complete spectrum, a distractor-only exchange gives a matched control. This paired setup brings a second question into focus: when does moving weight to the source help? The answer depends on the spectrum’s selective capacity, its assignment to task-labeled positions, and the downstream utility of the moved values.

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The controlled result is clear. Source-max raises per-token accuracy by 0.153, source-min lowers it by 0.043, and the distractor control changes it by 0.011. As the spectrum sharpens, the gap between the two assignments grows even though sharpening the original assignment remains near zero. A local utility term then predicts the finite margin effect with 0.917 Pearson correlation and 99.5% sign agreement. The same direction appears in Qwen and Pythia, while Gemma and SmoLLM2 reveal saturation and selector mismatch. Utility-selected routing also transfers to context-grounded natural QA.

Our contributions are:

- **Fixed-spectrum identification.** We separate selective capacity from task-conditioned assignment by preserving the complete attention spectrum and altering only its token placement.
- **Utility-conditioned mechanism.** We characterize when source–donor reassignment changes the final output and test the prediction against exact interventions.
- **Transfer and boundaries.** We validate the mechanism across controlled tasks, pretrained models, multi-evidence computation, and natural QA while identifying saturation and selector-mismatch regimes.

2 Related Work

Attention concentration and length generalization. Attention concentration has been connected to optimization, token selection, extrapolation, and sparse long-context computation (Tarzanagh et al. 2023; Wang et al. 2024; Nakanishi 2025; Vasylenko et al. 2025). Other accounts emphasize attention-output scale or normalization (Li, Boduljak, and Zhou 2025; Zsámboki et al. 2025). These statistics often move together as context length changes. That makes them useful clues, but it also makes causal interpretation difficult. Our intervention tests whether any one of them identifies task-correct routing while the entire spectrum is held fixed.

Attention intervention and non-identifiability. Attention weights have long been studied as explanations, perturbed to test prediction sensitivity, and shown to be non-identifiable without their value vectors (Jain and Wallace 2019; Wiegreffe and Pinter 2019; Brunner et al. 2020; Kobayashi et al. 2020; Hassid et al. 2022). Causal circuit analysis takes the

One attention spectrum, two token assignments

Query: Japan -> ? Required evidence: Tokyo

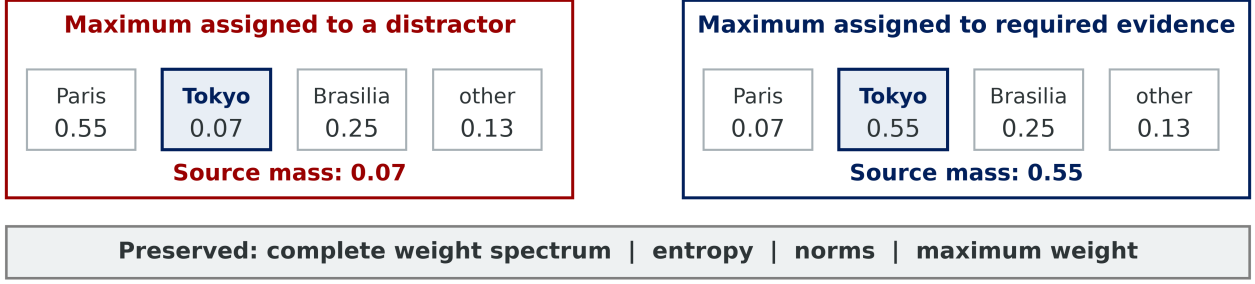


Figure 1: Fixed-spectrum reassignment changes destination while preserving concentration. The same four weights place the maximum on a distractor (left) or the required evidence (right). Output behavior can change only through the moved values and the downstream circuit that reads them.

next step by patching internal activations and measuring the output change (Yamakoshi et al. 2023; Conmy et al. 2023). Attribution patching uses a gradient approximation to screen many candidate components efficiently (Syed, Rager, and Conmy 2023; Kramár et al. 2024). Our utility term has a related first-order form. Here we evaluate it along an exact permutation of a model’s own attention row, alongside the finite swap itself.

Closest attention interventions. Attention Knockout blocks selected edges and renormalizes the row to trace factual information flow (Geva et al. 2023). Efficient attention projects a row onto its output-effective subspace and studies when attention supports causal explanation (Naim and Asher 2024). Contribution Weights combine attention mass with projected-value magnitude and alignment, then test token importance through removal (Cunningham and Muca Cirone 2026). These approaches delete, block, renormalize, or rank parts of an attention pathway. Our reassignment leaves the complete attention-weight multiset intact and changes its assignment to token positions. Equation 5 then connects that finite, task-labeled swap to the target margin.

Retrieval-based long-context architectures. Sparse and retrieval-based architectures ask how a model can access distant context efficiently (Vasylenko et al. 2025; Hu et al. 2025). Grouped Cross Attention, for example, learns to retrieve useful chunks into a bounded attention window. We study a different object: a trained attention row with a fixed weight spectrum. The goal is to isolate the behavioral effect of assigning its available selectivity to task-required evidence.

3 Methods

3.1 Capacity and Assignment

At an answer query, let $a = (a_1, \dots, a_n)$ be an attention row over n token positions, where $a_j \geq 0$ and $\sum_j a_j = 1$. Let $S \subseteq \{1, \dots, n\}$ contain the $r = |S|$ task-required evidence positions, and let $a_{(1)} \geq \dots \geq a_{(n)}$ denote the same weights

in descending order. Define

$$C_r^+ = \sum_{i=1}^r a_{(i)}, \quad C_r^- = \sum_{i=n-r+1}^n a_{(i)}, \quad (1)$$

$$K_r(a) = C_r^+ - C_r^-.$$

C_r^+ and C_r^- are the largest and smallest evidence masses available under permutations of the fixed spectrum, and K_r is the available routing range. An assignment is a permutation π of $\{1, \dots, n\}$ that places weight $a_{\pi(j)}$ at token position j . Its evidence mass and normalized assignment coordinate are

$$m_S(\pi) = \sum_{j \in S} a_{\pi(j)}, \quad (2)$$

$$\rho_S(\pi) = \frac{m_S(\pi) - C_r^-}{K_r(a)} \in [0, 1], \quad K_r(a) > 0.$$

Thus $m_S(\pi) = C_r^- + \rho_S(\pi)K_r(a)$. If $K_r(a) = 0$, every assignment has the same evidence mass.

Capacity specifies how much evidence mass the spectrum can support.
Assignment specifies where that mass is placed.
Utility specifies whether moving it changes the answer.

Proposition 1 (Fixed-spectrum capacity–assignment decomposition). *For every assignment π of a fixed weight multiset,*

$$C_r^- \leq m_S(\pi) \leq C_r^+, \quad (3)$$

and both bounds are attained. If $K_r(a) > 0$, Equation 2 defines a unique $\rho_S(\pi)$. No statistic invariant to token permutation can determine ρ_S .

Proof sketch. An r -position evidence set receives at most the r largest weights and at least the r smallest. Assigning those weights to S attains each endpoint. Normalizing evidence mass within this interval gives ρ_S . Any permutation-invariant statistic is constant across all assignments, including assignments at opposite endpoints. The supplement gives the full exchange argument and the corresponding affine margin range.

3.2 Utility-Conditioned Output Effects

Capacity and assignment describe attention mass. Prediction also depends on values and downstream computation. Let H be the selected attention heads. In each $h \in H$, let s be the evidence position, d_h the donor position whose weight is exchanged with it, and a_{hj} the original attention weight on position j . Define the transferred mass $\delta_h = a_{hd_h} - a_{hs}$. Let x_{hj} be token j 's residual-stream contribution after head h 's value and output projections. If z is the unpatched residual state, the intervention gives $z' = z + \Delta z$, where

$$\Delta z = \sum_{h \in H} \delta_h (x_{hs} - x_{hd_h}). \quad (4)$$

Let $\ell_k(z)$ be the downstream logit for token k , with target y and competitor c . Define $M_c(z) = \ell_y(z) - \ell_c(z)$, the exact change $\Delta M_c = M_c(z + \Delta z) - M_c(z)$, and local utility $u_{hj}^{(c)} = \nabla_z M_c(z)^\top x_{hj}$.

Proposition 2 (Utility-conditioned swap effect). *If $\nabla_z M_c$ is L_c -Lipschitz on $\{z + t\Delta z : t \in [0, 1]\}$, then*

$$\Delta M_c = \sum_{h \in H} \delta_h (u_{hs}^{(c)} - u_{hd_h}^{(c)}) + R, \quad |R| \leq \frac{L_c}{2} \|\Delta z\|_2^2. \quad (5)$$

For an affine downstream margin, $R = 0$.

The Taylor remainder is standard. Our contribution is the task-labeled capacity-assignment coordinate and its coupling to the exact fixed-spectrum swap vector in Equation 4.

The intervention transfers attention mass from a donor token to the source. It helps when the source contributes more positively to the target margin than the displaced donor, unless downstream curvature overwhelms that gain. Source-max can therefore succeed, saturate, or fail under the same attention-mass rule. The supplement derives Equation 5, extends it to evidence sets, and states a sufficient margin certificate.

4 Experimental Setup

Controlled tasks. Argmax retrieval presents key-value pairs and requests the value paired with the largest key. Flag retrieval presents values with exactly one marked position and requests that value. Both have one labeled source. Two-evidence modular sum marks two digits and requests their sum modulo ten, so both sources are required. Four-layer decoders are trained with NoPE and RoPE and four seeds per cell; evaluation reaches $50\times$ training length. An eight-layer, width-512 grid checks scale. The inverse temperature $\beta \in \{1, 2, 4\}$ changes attention capacity while each assignment comparison preserves the resulting sorted spectrum.

Pretrained and natural-language tasks. In-context key-value recall lists pairs followed by a query key; the paired value is the labeled source. We test Qwen2.5, Pythia, Gemma, and SmolLM2 across increasing pair counts. The selector audit compares circuits ranked on disjoint calibration data by source mass or by output-conditioned utility. Natural MCQA uses four-choice questions with one gold passage and unrelated passages. Questions must be correct with the full context and gold passage alone, but incorrect without context.

This filtering defines a model-specific context-dependent subset and is not intended to estimate average performance over the full dataset. Its length ladder freezes the four-passage circuit before forming nested 8-, 16-, and 32-passage contexts with additional unrelated passages.

Interventions and estimation. The distractor control swaps the largest and smallest non-source weights, preserving source mass while permuting the same attention row. For a source position s , let $d^+ = \arg \max_j a_j$, $d^- = \arg \min_j a_j$, and $D = \{j : j \neq s\}$ over valid tokens. Source-max swaps (s, d^+) and source-min swaps (s, d^-) . In controlled and pretrained key-value recall, the distractor control uses the maximum and minimum within D . Natural QA uses a displacement-matched control. With $\delta = a_{d^+} - a_s$, it chooses

$$(p, q) = \arg \min_{\{j, k\} \subset D} |a_j - a_k| - \delta \quad (6)$$

and swaps (p, q) . This matches the treatment's L_1 displacement up to the minimum attainable error while holding source mass fixed. Every control preserves the complete spectrum. The natural control matches displacement without matching the donor, positional distance, or value norm. Calibration and evaluation examples are disjoint, circuits are frozen before paired evaluation, and hierarchical intervals resample calibration or model seeds above paired examples. We interpret a cell only after its frozen competence gate passes.

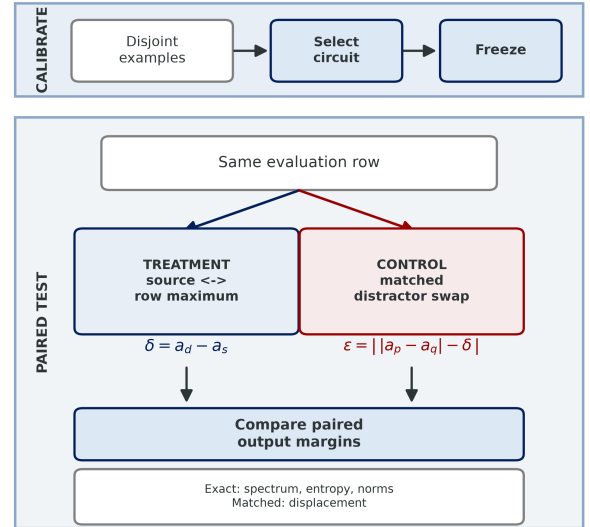


Figure 2: Paired causal protocol. A circuit selected on disjoint calibration data is frozen before treatment and control are applied to the same evaluation row. Both paths preserve the spectrum; the natural QA control additionally minimizes displacement mismatch in Equation 6.

Measured outcomes. We record task accuracy, target-versus-competitor margin, source mass, attention-output

variance, entropy, norms, and the full sorted spectrum. Every fixed-spectrum condition preserves the selected attention rows to numerical precision. The supplement reports prompts, hyperparameters, per-length tables, competence failures, and preregistered decision rules.

5 Results and Discussion

5.1 Assignment Changes Accuracy at Fixed Spectrum

We begin with the simplest test: move the largest available weight to the source, move it away, or exchange two distractors. Across sixteen trained models, the mean changes are +0.153, −0.043, and +0.011, respectively (Figure 3). Source-max beats the control in every model-length comparison, while complete-spectrum invariant error stays below 10^{-6} .

The effect grows with circuit coverage (Figure 3b). Patching one, two, four, and eight selected heads gives mean source-max gains of 0.007, 0.013, 0.056, and 0.153. The trajectory is monotone in 31 of 32 model-length comparisons. This rules out an isolated-head artifact and connects the single-row proposition to a distributed retrieval circuit.

The baseline length ladder helps explain this gap. Source attention tracks accuracy more closely than attention-output variance. Normalizing that variance leaves accuracy unchanged, whereas forcing mass onto the known source restores controlled retrieval. Sharpening the original row has little effect. Taken together, these checks indicate that task-correct routing drives the effect; global scale, entropy, and variance do not account for it.

The ordering survives the eight-layer replication, both positional encodings, four training seeds, and two controlled tasks. Random-layer and adjacent-layer interventions remain near zero when the selected circuit is held fixed. Full model-level intervals and the variance and patching results appear in the supplement.

5.2 Capacity Amplifies Assignment

We next vary spectrum and assignment independently. Temperature scaling changes the maximum available weight, and at every temperature we evaluate identity, source-max, source-min, and distractor control on the same examples. The source-max minus source-min contrast grows from 0.196 to 0.346, while sharpening the identity assignment changes accuracy by at most 0.009 (Figure 4a). In other words, capacity sets the possible leverage, while assignment determines where that leverage lands.

For held-out examples we differentiate the target margin along the exact source-max path. The first-order term in Proposition 2 predicts the finite margin change with 0.917 Pearson correlation, 0.989 Spearman correlation, and 99.5% sign agreement (Figure 4b). The remaining error measures downstream curvature along the swap.

Modular sum requires the model to route and combine two marked values. At $5\times$ training length, assigning the top two weights to evidence raises exact match by 0.066, assigning the bottom two lowers it by 0.291, and evidence-max exceeds a control that preserves both the spectrum and total evidence

mass by 0.043 (95% CI [0.017, 0.072]). The result therefore extends beyond copying one token. A registered three-evidence estimate includes zero, and four-evidence models miss the frozen competence gate. Higher arity remains unresolved.

Evidence-max and evidence-min preserve the complete spectrum. The multi-evidence control also preserves total evidence mass, changing only which evidence token receives each weight. Its small effect separates aggregate source mass from useful evidence assignment.

5.3 The Same Intervention Produces Three Pretrained Regimes

The pretrained models show why the utility term matters. Qwen and Pythia gain margin when source-max replaces the distractor control. Gemma stays accurate as context grows and gains little from extra source mass, even though source-min reliably harms it. This is the saturation regime predicted by Proposition 2: removing evidence matters, while extra routing has little left to add. A quantized Qwen2.5-7B run preserves the positive margin direction.

SmolLM2 provides the complementary case. Its preregistered source-mass selector fails, yet ranking heads on disjoint calibration examples by output-conditioned gradient or utility restores a positive effect. A ten-seed ablation ranks source gradient, utility gain, and utility gap above source mass and transfer mass. Rankings vary by architecture, but the broader lesson holds: circuit selection must account for the output.

Interpolating from the natural row to the utility-selected swap gives a monotone margin path over five calibration seeds. On context-grounded natural QA, utility-selected routing improves margin over a displacement-matched control, while source-mass selection harms it. An unconstrained generation audit finds no first-answer or repetition cost.

The Qwen circuit-selection audit varies calibration seed, head budget, and layer. All nine selected-circuit cells at $K \in \{2, 4, 8\}$ have positive max-control margin effects, and all nine paired intervals exclude zero. Their mean effect is +0.874, compared with −0.059 for random-layer circuits. Adjacent-layer effects are smaller and inconsistent, which localizes the result to the selected circuit. The first-order utility audit also passes independently for Qwen, Pythia, and Gemma across three calibration seeds per family. A quantized Qwen2.5-7B replication retains the positive direction. Complete cells and intervals appear in the supplement.

5.4 Natural Context Exposes the Boundary of the Mechanism

The fixed-context natural-QA result establishes causal routing in ordinary text. The length ladder asks a different question: does the same mechanism explain degradation as context grows? We select the circuit at four passages, freeze every choice, and append unrelated passages to form nested 4, 8, 16, and 32-passage contexts. Hierarchical intervals resample calibration seeds and paired questions, so repeated observations from one circuit are not treated as independent models.

Both Qwen and SmolLM2 lose margin and accuracy by 32 passages, yet their source-mass trajectories diverge. Qwen

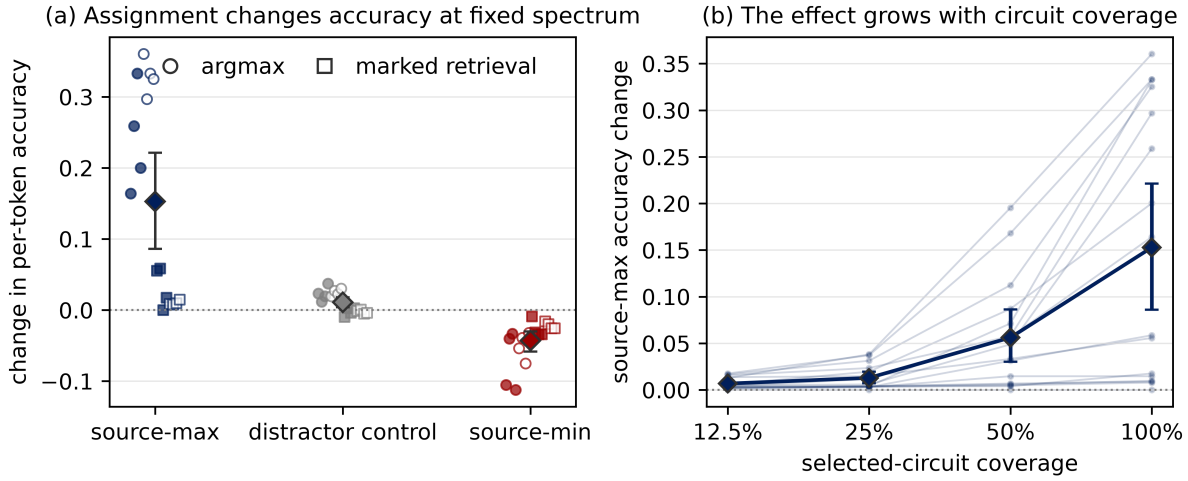


Figure 3: Assignment changes retrieval despite an identical attention spectrum. Source-max improves accuracy, source-min harms it, and the distractor-only control remains near zero (a). The effect increases with the fraction of the selected circuit receiving the intervention (b). Diamonds and intervals summarize trained models; small points and faint paths show individual cells.

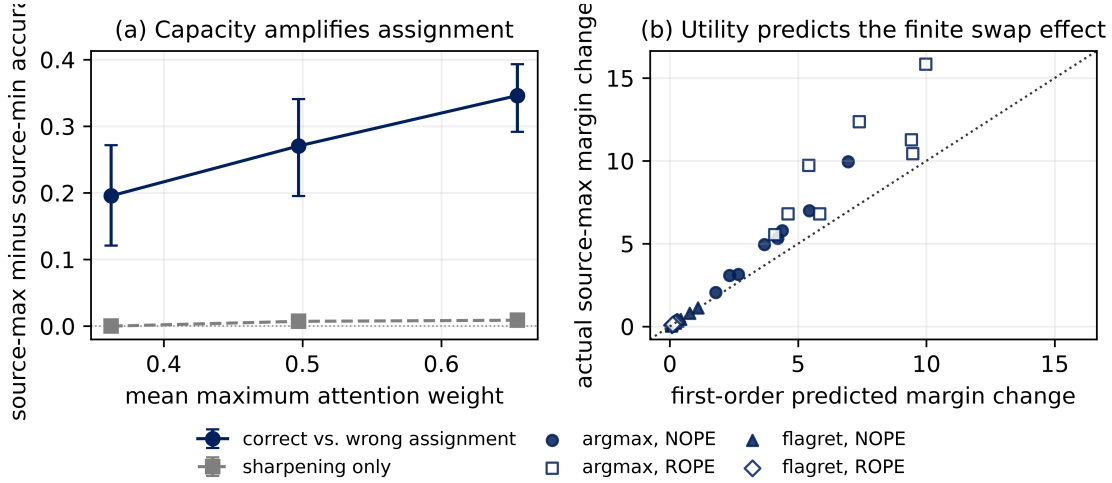


Figure 4: Capacity controls leverage, and utility predicts the finite intervention. Sharper spectra increase the correct-versus-wrong assignment contrast while sharpening alone remains near zero (a). The first-order utility term predicts the exact source-max margin effect across tasks and positional encodings (b). Error bars summarize trained models.

source mass rises as intervention rescue weakens. SmoLM2 source mass falls while rescue stays flat. The shared degradation therefore comes from more than one source-mass trajectory.

This result marks the boundary of the assay. At a fixed context, a spectrum-preserving swap measures whether a chosen circuit can use reassigned evidence. As contexts grow, representation quality, distractor semantics, answer competition, and computation outside that circuit can also matter. The theorem therefore concerns a specified swap at a fixed context, while the full set of long-context bottlenecks remains open.

The competence checks make the remaining boundary equally clear. In the arity study, the two-evidence task passes and supports the mass-preserving control; the three-evidence

interval includes zero, and the four-evidence model misses competence. Higher arity remains unresolved. An eight-token generation audit also leaves first-answer accuracy and repetition unchanged, so simple format collapse is not driving the result.

5.5 Identification and Replication Audits

We then ask whether the result depends on one calibration sample, patch budget, or model family. Across three frozen Qwen calibration seeds, selected circuits are positive at every head budget. Adjacent-layer effects are smaller and inconsistent, while random layers average near zero (Table 3).

The utility relation also survives a separate replication check. Across three seeds per family, every competent

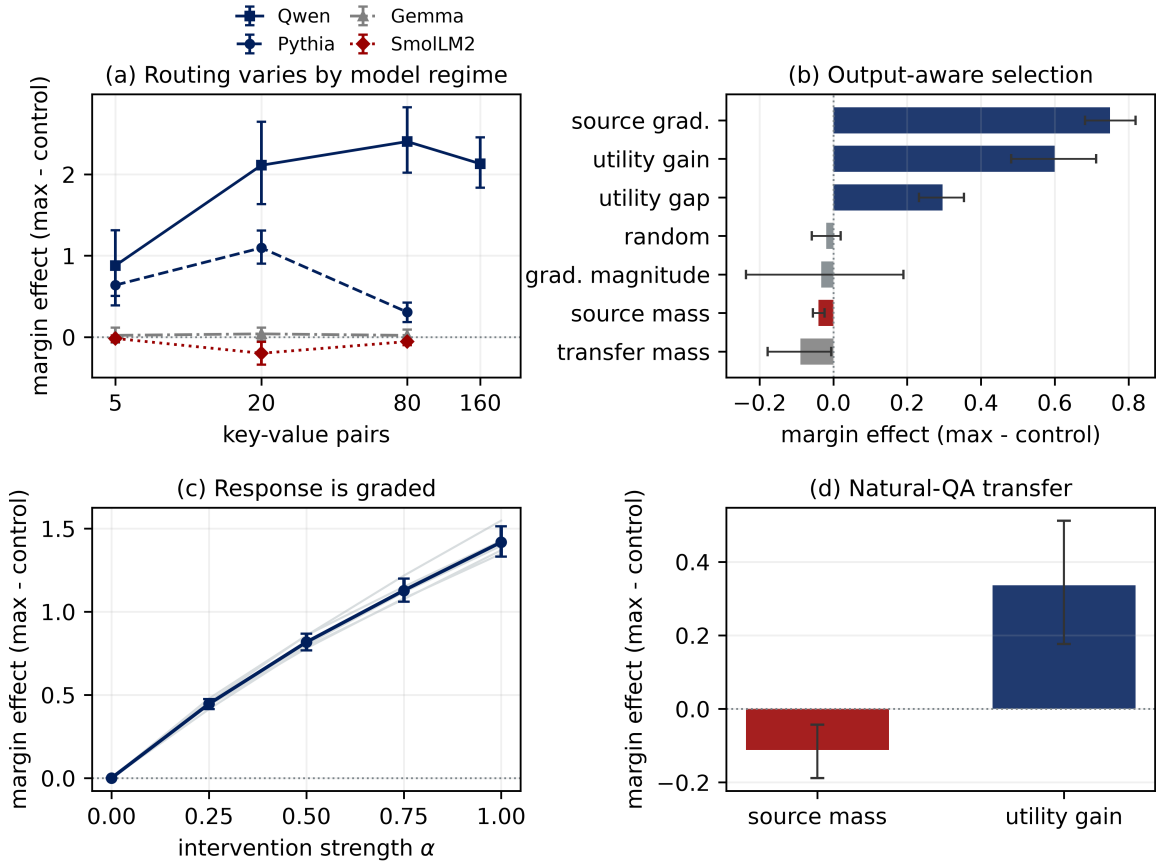


Figure 5: Fixed-spectrum routing transfers conditionally across pretrained models. Qwen and Pythia are routing-responsive in the selected circuit, Gemma is saturated, and source-mass selection mismatches the useful SmoLLM2 circuit (a,b). The repaired effect is graded (c) and transfers to natural QA (d). Error bars show 95% intervals; faint lines in (c) are independent calibration seeds.

Fixed-spectrum assignment		n	Unpatched	Δ source-max	Δ source-min	Δ distractor	max – distractor
Single evidence, accuracy	$\beta = 1$	8,192	.619	+ .153	– .043	+ .011	+ .142
	$\beta = 4$	8,192	.619	+ .298	– .048	+ .025	+ .273
Two evidence, exact match	$1 \times$	2,048	.952	+ .019	– .559	+ .012	+ .007
	$5 \times$	2,048	.484	+ .066	– .291	+ .023	+ .043
Pretrained KV, margin	Pythia-1.4B	384	– 3.805	+ .826	– 1.022	+ .146	+ .680
	Qwen2.5-1.5B	512	– 4.035	+ 2.044	– .758	+ .162	+ 1.882
	Qwen2.5-7B	288	– 3.361	+ .623	– .907	+ .055	+ .568
	Gemma-2-2B	384	+ 2.234	+ .040	– 1.373	+ .013	+ .027
Output-conditioned circuit selection		n	ΔM versus matched control		ΔM source mass	$\Pr(\Delta M > 0)$	utility – source
Task	Model		Utility gain	Utility gap			
Pretrained KV	Pythia-1.4B	384	+ .519	+ .359	+ .006	.810	+ .513
	SmoLLM2-1.7B	384	+ 1.253	+ .859	+ .006	.969	+ 1.247
	Qwen2.5-1.5B	384	+ 1.493	+ 1.493	+ .069	.956	+ 1.424

Table 1: Benchmark summary with direct column semantics. The upper block reports changes from the unpatched model and highlights the treatment-minus-control contrast. The lower block reports source-max margin effects for circuits chosen by each selector, each measured against its fixed-spectrum control. $\Pr(\Delta M > 0)$ is the fraction of paired evaluation examples with a positive utility-gain source-max minus matched-control margin. Bold final columns remain identifiable in grayscale. Full cells and 95% intervals are in the supplement.

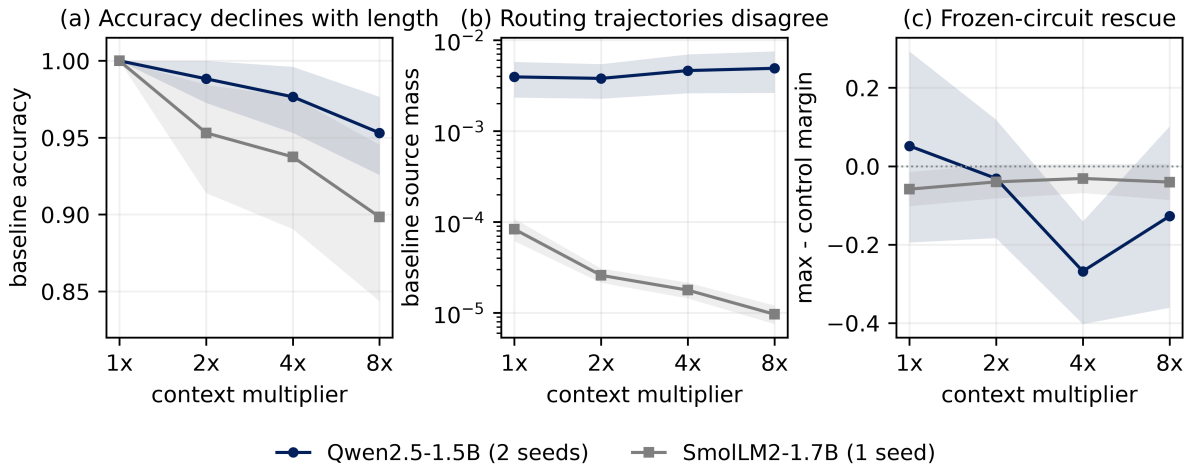


Figure 6: Natural-QA degradation follows divergent routing trajectories. Circuits are frozen from four through 32 passages, shown as $1\times$ – $8\times$ context. Accuracy falls (a), source routing moves in opposite directions (b), and rescue stays flat or weakens (c). Bands are hierarchical 95% intervals.

Model	Δmargin	$\Delta\text{acc.}$	Δm_S	Δrescue
Qwen2.5-1.5B	-1.036	-.047	+ .00095	-.179
SmolLM2-1.7B	-.193	-.102	-.00007	+ .018

Table 2: Natural-context degradation follows divergent source-mass trajectories. Changes compare 32 with four passages using circuits frozen at four.

Circuit	Mean ΔM	Hierarchical 95% CI	Seed SD
Selected, $K = 2$	+.460	[+.233,+.723]	.180
Selected, $K = 4$	+1.054	[+.652,+1.493]	.346
Selected, $K = 8$	+1.107	[+.739,+1.519]	.293
Adjacent -1	+.147	[-.002,+.332]	.155
Adjacent +1	-.131	[-.390,+.211]	.323
Random layer	-.059	[-.207,+.101]	.151

Table 3: The selected Qwen circuit is stable across seeds and head budgets. Intervals resample calibration seeds and paired examples; SD is across seedwise mean effects.

Model	Mean Spearman	Seed range	Sign agreement
Qwen2.5-1.5B	.866	[.832,.898]	.805
Pythia-1.4B	.544	[.500,.586]	.807
Gemma-2-2B	.554	[.534,.582]	.714

Table 4: Local utility predicts exact source-max effects across pretrained families. Means and ranges summarize three disjoint calibration seeds.

source-max cell is positive, and predicted effects rank the exact effects within each cell (Table 4).

Output-conditioned selectors lead in each family, although their internal ranking changes by architecture. Full selector results and competence-gate failures appear in the supple-

ment.

Supplementary audits cover invariants, controls, calibration, and competence gates. The natural length mechanism remains open.

6 Limitations

- Evidence positions must be identifiable. Source-max uses labels and serves as a causal probe; it is unsuitable as a deployable inference algorithm.
- Utility selection needs disjoint task-conditioned calibration and varies across model families.
- Higher evidence arity remains inconclusive or competence-limited; order-dependent computation is outside the current intervention.
- Natural long-context failure has multiple causes. We make no universal source-mass-loss claim.

7 Conclusion

Attention concentration tells us how much selectivity is available, while assignment tells us where that selectivity goes. Our capacity–assignment decomposition and utility term identify when a fixed-spectrum swap changes the target margin. This gives a local claim about the patched circuit.

The assay reproduces across controlled retrieval, two-evidence computation, and pretrained models. Qwen and Pythia respond to reassignment in the selected circuits, Gemma is saturated, and SmolLM2 benefits from output-conditioned circuit selection. Dose response, disjoint calibration, layer controls, and seedwise replication support these distinctions.

Natural QA carries the assay into ordinary language at a fixed context. The length ladder then marks its boundary: accuracy can decline even when source mass and intervention rescue follow different trajectories. The assay diagnoses a specified routing event and leaves the full account of long-context failure open.

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